Laporan Praktikum PBO: Composition & Inheritance

Mata Kuliah: Pemrograman Berorientasi Objek / Kelas A
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Topik: Pertemuan 6 - Latihan Inheritance (Latihan 1, 2, dan 3)

1. Latihan 1: Komposisi (Line Composed by 2 Points)

Relasi *has-a*: Class Line memiliki 2 atribut objek Point (titik awal begin dan titik akhir end). Line mendelegasikan perhitungan koordinat kepada objek Point.

```
public class Line {
    private Point begin;
    private Point end;
    public Line(int x1, int y1, int x2, int y2) {
        this.begin = new Point(x1, y1);
        this.end = new Point(x2, y2);
    }
    public double getLength() {
        int xDiff = end.getX() - begin.getX();
        int yDiff = end.getY() - begin.getY();
        return Math.sqrt(xDiff * xDiff + yDiff * yDiff);
    }
}
```

2. Latihan 2: Pewarisan (LineSub extends Point)

Relasi *is-a*: Class LineSub memperluas (extends) Point sebagai titik awalnya (koordinat superclass), dan menambahkan atribut end (Point) sebagai titik akhirnya.

```
public class LineSub extends Point {
    private Point end;
    public LineSub(int x1, int y1, int x2, int y2) {
        super(x1, y1);
        this.end = new Point(x2, y2);
    }
    public double getLength() {
        int xDiff = end.getX() - super.getX();
        int yDiff = end.getY() - super.getY();
        return Math.sqrt(xDiff * xDiff + yDiff * yDiff);
    }
}
```

3. Latihan 3: Pewarisan (Point3D extends Point)

Class Point3D memperluas Point untuk titik 3 dimensi dengan menambahkan koordinat z dan meng-override method toString().

```
public class Point3D extends Point {
    private int z;
    public Point3D(int x, int y, int z) {
        super(x, y);
        this.z = z;
    }
    public int getZ() { return z; }
    public void setZ(int z) { this.z = z; }
    @Override
    public String toString() {
        return "(" + super.getX() + ", " + super.getY() + ", " + z + ")";
    }
}
```

4. Output Pengujian Lengkap (TestAll.java)

```
>>> [LATIHAN 1: Composition (Line has two Points)] <<<
Representasi line1 : Line[begin=(1, 2), end=(4, 6)]
Panjang line1      : 5.0 (Harusnya 5.0)

>>> [LATIHAN 2: Inheritance (LineSub extends Point)] <<<
Representasi lineSub1: LineSub[begin=(2, 3), end=(5, 7)]
Panjang lineSub1     : 5.0 (Harusnya 5.0)

>>> [LATIHAN 3: Inheritance (Point3D extends Point)] <<<
Point3D berparameter (10, 20, 30):
Representasi p3d: (10, 20, 30)
Koordinat X (superclass): 10, Y (superclass): 20, Z (subclass): 30
```